

COMPUTATIONAL

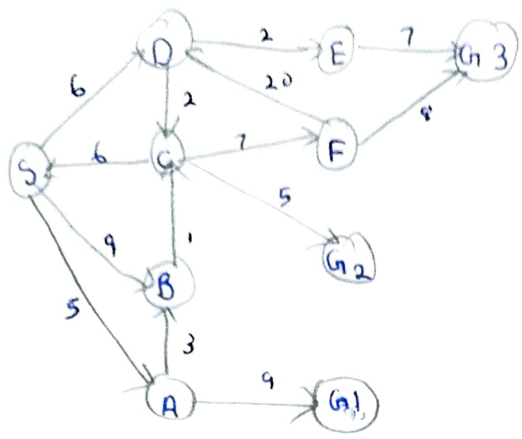
INTELLIGENCE

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DEPTH FIRST SEARCH:

Stack

dfs(S)

dfs(E)

dfs(D)

dfs(S)

next[]:

[S]

[S D E]

Stack

dfsRec(D)

dfsRec(S)

dfs(G3) ^{Found}

dfs(E)

dfs(D)

dfs(S)

Res[]:

[S D]

[S D E G3]

Path: S → D → E → G3

BREADTH FIRST SEARCH

res[]: [S]

queue: [D B A]

res[]: [S D]

queue: [B A E C]

res[]: [S D B]

queue: [A E C]

res[]: [S D B A]

queue: [E C G1]

Path: S → D → B → A → E → C → G1

UNIFORM COST SEARCH :

[S]

S [A, D, B]

A [D, B, G₁]

D [C, E, B, G₁]

C [E, B, G₂, G₁, F]

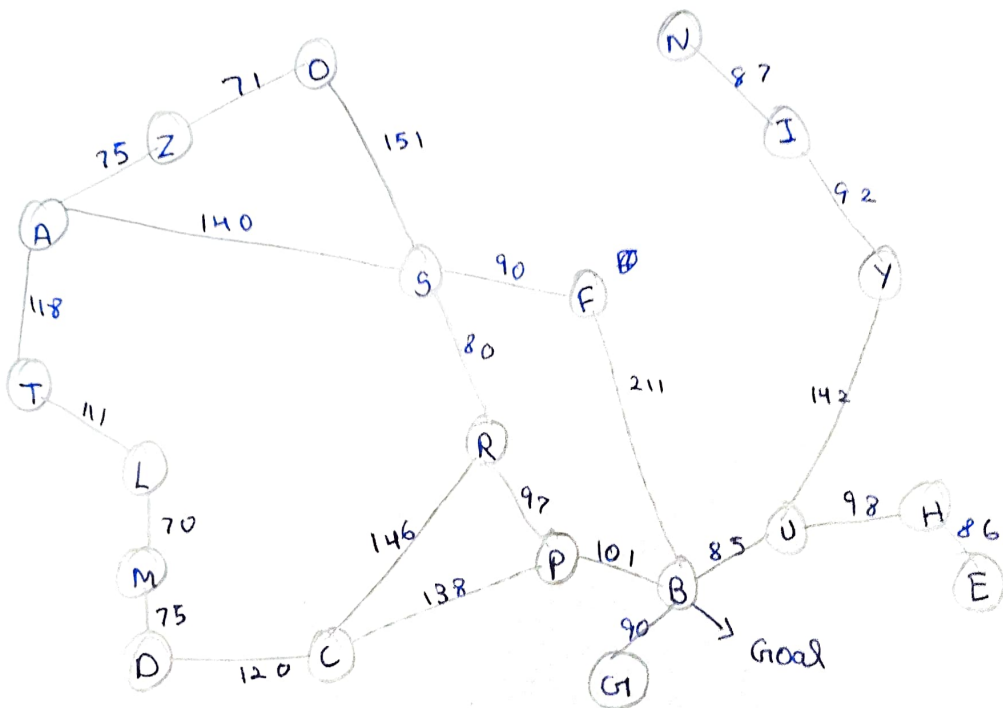
E [B, G₂, G₁, G₃, F]
 ↳ minimal cost

UCS Path :

S → D → C → G₂

⇒ 6 + 2 + 5

⇒ 13



Stack :

res[] :

dfs(A)

A

dfs(Z)

A Z

dfs(A)

Stack

res[] :

dfs(O)

A Z O

dfs(Z)

dfs(A)

dfs(S)
dfs(O)
dfs(Z)
dfs(A)

res[]:

A Z O S

dfs(F)
dfs(S)
dfs(O)
dfs(Z)
dfs(A)

res[]:

A Z O S F

dfs(B)
dfs(F)
dfs(S)
dfs(O)
dfs(Z)
dfs(A)

A Z O S F B

DFS Path:

A → Z → O → S → F → B

BREADTH FIRST SEARCH:

res[] = A

queue = Z S T

res[] = A Z

queue = S T O

res[] = A Z S

queue = T O F R

res[] = A Z S T

queue = O F A L

res[] = A Z S T O

queue = F R L

res[] = A Z S T O F

queue = R L B

BFS Path:

A → Z → S → T → O → F → R → L → B

UNIFORM COST SEARCH:

$[A]$
 $A \begin{matrix} 75 & 118 & 140 \\ [2, T, S] \end{matrix}$
 $2 \begin{matrix} 118 & 140 & 146 \\ [T, S, 0] \end{matrix}$
 $T \begin{matrix} 140 & 146 & 229 \\ [S, 0, L] \end{matrix}$
 $S \begin{matrix} 146 & 220 & 229 & 230 \\ [0, R, L, F] \end{matrix}$

$\downarrow$ [NO Path]

Backtrack

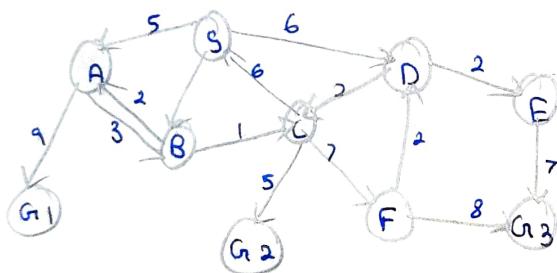
$\begin{matrix} 229 & 230 & 317 & 366 \\ R [L, F, P, C] \end{matrix}$
 $\begin{matrix} 230 & 299 & 317 & 366 \\ L [\cancel{A}, M, P, C] \end{matrix}$
 $\begin{matrix} 299 & 317 & 366 & 441 \\ F [M, P, C, B] \end{matrix}$
 $\begin{matrix} 317 & 366 & 374 & 441 \\ M [P, C, D, B] \end{matrix}$
 $\begin{matrix} 366 & 374 & 418 & 441 \\ P [C, D, B, B] \end{matrix}$
 $\downarrow$
 min cost

U.S.G Path:

$A \rightarrow S \rightarrow R \rightarrow P \rightarrow B$

$\Rightarrow 140 + 80 + 97 + 101$

$= 418$



Stack:

dfs(s)

res[]:

S

stack

dfs(s)
 dfs(A)
 dfs(s)

Res[]:

S A G1

dfs(A)

dfs(s)

S A

DFS Path: S A $\rightarrow$ G1

BREADTH FIRST SEARCH:

Stack:

res[]:

S	
---	--

queue:

A	B	D
---	---	---

res[]:

S	A
---	---

queue:

B	D	G1
---	---	----

BFS Path: $S \rightarrow A \rightarrow B \rightarrow D \rightarrow G1$

UNIFORM COST SEARCH:

[S]

⁵
S [A, D, B]

⁶ ⁸ ⁹ ¹⁹
A [D, B, B, G1]

⁸ ⁸ ⁸ ⁹ ¹⁰
D [B, E, C, B, G1]

⁸ ⁸ ⁹ ⁹ ¹⁰ ¹⁹
B [E, C, B, C, A, G1]

⁸ ⁹ ⁹ ¹⁰ ¹⁴ ¹⁵
E [C, B, C, A, G1, G3]

⁹ ⁹ ¹⁰ ¹² ¹⁴ ¹⁴ ¹³ ¹⁵
C [B, C, A, G2, G1, S, G3, F]

min cost

UES Path:

$S \rightarrow D \rightarrow C \rightarrow G2$

$\Rightarrow 6 + 2 + 5$

$\Rightarrow 13$